5.13.40

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Question

Let $a, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations

$$\begin{aligned} ax + 2y &= \lambda \\ 3x - 2y &= \mu \end{aligned}$$

Which of the following statement(s) is (are) correct?

1. If $a = -3$, then the system has infinitely many solutions for all value of λ and μ.
2. If $a \neq -3$, then the system has unique solution for all values of λ and μ.
3. If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$.
4. If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$.

Solution

By the system of equations,we get the following augmented matrix:

$$\left(\begin{array}{cc|c} a & 2 & \lambda \\ 3 & -2 & \mu \end{array}\right) \xrightarrow{R_1 \longrightarrow R_1 + R_2} \left(\begin{array}{cc|c} a+3 & 0 & \lambda+\mu \\ 3 & -2 & \mu \end{array}\right) \tag{1}$$

Now if $a = -3$,
Case-1: If $\lambda + \mu = 0$ and the system has **infinitely many solutions**.
Case-2: If $\lambda + \mu \neq 0$, then the system has **no solution**

and if $a \neq -3$, then the first row is non-zero and matrix is invertible.
Hence system has a **unique solution** for all values of λ and μ.

Solution

and if $a \neq -3$, then the first row is non-zero and matrix is invertible. Hence system has a **unique solution** for all values of λ and μ.

And solution for $a \neq -3$ is:

$$x\left(a+3\right) = \lambda + \mu \implies x = \frac{\lambda+\mu}{a+3} \tag{2}$$

$$3x - 2y = \mu \implies y = \frac{3\lambda - a\mu}{2a+6} \tag{3}$$

Therefore (2),(3) and (4) are the correct statements.

C Code

```c
#include <stdio.h>

void solveSystem(double a, double lambda, double mu, double sol[2]) {
    double D = -2 * (a + 3);

    if (D != 0) {
        sol[0] = (-2 * lambda - 2 * mu) / D; // x
        sol[1] = (3 * lambda - a * mu) / D; // y
        printf(Unique solution exists:\n);
        printf(x = %.3f, y = %.3f\n, sol[0], sol[1]);
    } else {
        // When a = -3
        if (lambda + mu == 0)
            printf(Infinitely many solutions (system consistent)\n);
        else
            printf(No solution (system inconsistent)\n);
    }
```

```
import ctypes

# Load shared library
lib = ctypes.CDLL(./code.so)

# Define argument and return types
lib.solveSystem.argtypes = [
    ctypes.c_double, ctypes.c_double, ctypes.c_double,
    ctypes.POINTER(ctypes.c_double)
]
```

Python Code

```python
# Prepare result array
sol = (ctypes.c_double * 2)()

# ---- Fixed values ----
a = -3
lam = 2
mu = -2

# ---- Call C function ----
lib.solveSystem(a, lam, mu, sol)

# ---- Display result (if unique) ----
if a != -3:
    print(fx = {sol[0]:.3f}, y = {sol[1]:.3f})
```